

Incident Ticket #2: Complete Outage of Doctors Office Subnet

Status: Resolved

Skill Evaluation: Structured diagnostic process; identified STP and 802.1Q encapsulation mismatches across multiple layers

Ticket Overview

- Ticket ID: TCK-002
- Source Subnet: VLAN 25 (Doctors_Data) | Range: 192.168.25.10 - 11/24
- Primary Destination: VLAN 25 default gateway — 192.168.25.1
- Secondary Test Destinations: Reception VLAN 10 and Admin VLAN 20
- Symptom: Doctor office PCs cannot reach default gateway or other internal subnets. IP phones on VLAN 30 continue to function.
- Expected Behavior: Doctor PCs should reach their VLAN 25 default gateway and permitted internal VLANs via ROAS.
- Actual Behavior: 100% packet loss to local gateway and target subnets.

Diagnostic Matrix & Root Causes

Level	Component	Discovered Condition	Impact	Corrective Action
Host	PC Gateway	Set to 192.168.10.1 instead of 192.168.25.1	Packets routed to wrong subnet	Updated IP config to 192.168.25.1
Layer 2	Switch C Trunk (Fa0/3)	VLAN 25 missing from allowed list	VLAN 25 traffic dropped at trunk	Added VLAN 25 to switchport trunk allowed vlan
Layer 2	Switch C (Fa0/4)	Port in STP Broken state (BKN *PVID_Inc)	Native VLAN/PVID	Reconfigured port as access

			mismatch blocked traffic	VLAN 25 / Voice VLAN 30
Layer 3	ROAS Subinterface	FastEthernet0/0.25 set to encapsulation dot1Q 35	Router tagged frames for wrong VLAN	Re-encapsulated subinterface to VLAN 25

Step-by-Step Configuration Fixes

1. Switch C Trunk Update:

```
Switch-C(config)# interface FastEthernet0/3
Switch-C(config-if)# switchport mode trunk
Switch-C(config-if)# switchport trunk allowed vlan 10,20,25,30,40,50,60
```

2. Switch C Access Port Restoration:

```
Switch-C(config)# interface FastEthernet0/4
Switch-C(config-if)# switchport mode access
Switch-C(config-if)# switchport access vlan 25
Switch-C(config-if)# switchport voice vlan 30
Switch-C(config-if)# spanning-tree portfast
```

3. Router Inter-VLAN Subinterface Correction:

```
ROAS(config)# interface FastEthernet0/0.25
ROAS(config-subif)# encapsulation dot1Q 25
ROAS(config-subif)# ip address 192.168.25.1 255.255.255.0
```

Verification Proof

```
C:\> ping 192.168.25.1
```

```
Pinging 192.168.25.1 with 32 bytes of data:
Reply from 192.168.25.1: bytes=32 time<1ms TTL=255
Reply from 192.168.25.1: bytes=32 time<1ms TTL=255
Reply from 192.168.25.1: bytes=32 time<1ms TTL=255
Reply from 192.168.25.1: bytes=32 time<1ms TTL=255
```

Ping statistics for 192.168.25.1:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)

(Inter-VLAN connectivity confirmed from Reception VLAN 10 and Admin VLAN 20 to Doctors VLAN 25).